

Pressure in fluids and in the atmosphere

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 6 starred (★) questions first.

	QUESTION	ANSWER	✓
1	Name the two states of matter that are considered fluids.	Liquid and gas	☐
2	Describe what causes pressure in a fluid.	The particles of the fluid colliding with surfaces.	☐
3	A fluid exerts a force on its container. State the direction of the force.	At right angles to the surface.	☐
4	State the equation linking pressure, area and force.	pressure = force ÷ area ($p = F \div A$)	☐
5 ★	State the standard unit of pressure.	Pascal (Pa)	☐
6 ★	Describe how the pressure in a liquid depends on the height of the liquid column.	The taller the column, the greater the pressure.	☐
7 ★	Describe how the pressure in a liquid depends on its density.	The denser the liquid, the greater the pressure.	☐
8 ★	Explain what causes upthrust.	Pressure increases with depth, so the pressure on the bottom of a submerged object is greater than on the top.	☐
9	Describe the Earth's atmosphere.	A thin layer of air around the Earth.	☐
10	Describe how the density of the atmosphere changes with height.	It decreases as height increases.	☐
11	Give the cause of atmospheric pressure.	Air molecules colliding with a surface.	☐
12 ★	Describe how atmospheric pressure changes with height.	It decreases as height increases.	☐

QUESTION		ANSWER	✓
13	State the equation for the pressure due to a column of liquid.	pressure = height × density × gravitational field strength ($p = h \rho g$)	☐
14 ★	Explain why the pressure in a liquid increases with depth.	The deeper you go, the greater the weight of liquid above.	☐
15	State the condition for an object to float.	Upthrust equals its weight.	☐
16	State when an object in a liquid sinks.	When its weight is greater than the upthrust.	☐
17	Explain why atmospheric pressure decreases as height increases.	There is less air above, so fewer air molecules collide with each unit of area.	☐